Republic of the Philippines
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ITEP 414 – SYSTEM ADMINISTRATION AND MAINTENANCE

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PART III: BIOS VS UEFI COMPARISON

FEATURE	BIOS	UEFI
Definition	Basic Input/Output System — firmware stored on a chip that initializes hardware before the OS loads	Unified Extensible Firmware Interface — modern replacement for BIOS with a richer feature set
Boot Process	Reads the Master Boot Record (MBR), loads bootloader in 16-bit real mode	Reads the GUID Partition Table (GPT), can boot directly in 32/64-bit mode
Max Disk Support	2TB (due to MBR limitations)	Over 9 zettabytes (uses GPT)
Partition Style	MBR (max 4 primary partitions)	GPT (up to 128 partitions)
Security Features	None built-in	Secure Boot (blocks unsigned/malicious bootloaders)
Speed	Slower boot (sequential hardware checks)	Faster boot (parallel initialization)

UEFI is replacing BIOS, as it addresses the fundamental problems of BIOS. **BIOS** starts in 16-bit mode and only addresses 1MB of memory during start up; UEFI is in 32/64-bit mode and accesses memory directly, initializing the system much faster. BIOS uses MBR partitioning and supports drives up to 2TB which is a big issue when storage capacity started increasing. GPT is used by UEFI, completely eliminating that limit and giving way to many more partitions. Another key reason for security is that UEFI's Secure Boot mechanism ensures that only trusted, signed bootloaders and operating system kernels are loaded, whereas BIOS does not have any defense against rootkits and bootkits. UEFI also provides a graphical and mouse-driven configuration screen in place of BIOS's text-based menus, as well as capabilities such as network booting and quicker POST (Power-On Self-Test). As a result of these performance, capacity and security benefits, almost all Enterprise servers and modern PCs are now shipped with UEFI and BIOS mode is reserved for legacy support.